Rachel D. Howard, PhD

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Citizenship: USA | Pronouns: she/her/hers

Education

December 2024	**Ph.D. Soil, Water, and Ecosystem Sciences** *University of Florida	GPA: 3.78* Dissertation: Exploring The Impact of Hypoxia on Taxonomic Diversity and Functional Dynamics in Coral Microbiomes
May 2019	**BS Marine Biology**	Roger Williams University *GPA: 3.66*
December 2016	**AA Biology**	Nashua Community College *GPA: 3.70*

Research & Professional Experience

Postdoctoral Research Associate 2025-Present

University of Texas at Arlington– Department of Biology
Advisor: Dr. Laura Mydlarz

- **Lead 'omics integration efforts** as a RISE 100 postdoctoral associate at University of Texas Arlington.
- **Analyze bulk RNA-seq data** to characterize gene-expression responses to environmental stress and disease.
- **Develop reproducible bioinformatics workflows** for transcriptomic and microbiome data processing, normalization, statistical analysis, and visualization.
- **Integrate** host transcriptomic, algal symbiont transcriptomic, microbiome, and clinical datasets using **multiblock methods in the mixOmics R package**.
- **Apply DIABLO and sparse partial least-squares discriminant analysis** to identify correlated multi-omic signatures and candidate biomarkers.
- **Train, tune, and evaluate supervised machine-learning models** using cross-validation, feature selection, balanced error rates, confusion matrices, and ROC–AUC.
- **Perform functional annotation and pathway-enrichment analyses** to interpret genes and molecular pathways associated with disease and environmental stress.
- **Create publication-quality visualizations** of high-dimensional data, including heatmaps, ordinations, feature-loading plots, and correlation networks.
- **Translate complex multi-omic results** into biologically meaningful hypotheses about host–microbe interactions and disease susceptibility.
- **Collaborate with interdisciplinary researchers** to guide experimental design, data integration, statistical analysis, and interpretation.
- **Develop training materials and provide mentorship** in R, RNA-seq analysis, multi-omics integration, and reproducible computational workflows.

Technical Skills:

- **Programming & Computing**: R, Python, Bash, Unix/Linux, Git/GitHub, high-performance computing (HPC), R Markdown
- **Bioinformatics**: RNA-seq, 16S rRNA amplicon sequencing, metatranscriptomics, multi-omics integration, differential expression, functional annotation, pathway enrichment
- **Methods**: Machine learning, supervised classification, feature selection, cross-validation, sPLS-DA, DIABLO, PERMANOVA, ordination, network analysis
- **Software & Packages**: mixOmics, DESeq2, DADA2, phyloseq, vegan, clusterProfiler, Trinity, Salmon, Cytoscape
- **Visualization**: ggplot2, ComplexHeatmap, multivariate visualization, correlation networks, publication-quality figure development
- **Research & Leadership**: Experimental design, reproducible workflow development, interdisciplinary collaboration, scientific communication, bioinformatics training and mentorship

Postdoctoral Scholar 2024-2025

Texas State University – Department of Biology
Advisor: Dr. Lauren Fuess

- **Lead bioinformatics research efforts** as part of the NSF CREST Center Planning Grant, supporting the establishment of a new research center focused on global change bioinformatics.
- **Develop and implement bioinformatics methodologies** to analyze large-scale genomic and transcriptomic datasets related to climate change impacts on marine organisms.
- **Curate and manage high-throughput sequencing datasets**, ensuring efficient data processing, quality control, and accessibility for faculty and student researchers.
- **Train and mentor students and faculty** in bioinformatics techniques, including high-performance computing (HPC), command-line tools and R/Python-based data analysis, fostering cross-disciplinary collaborations.
- **Collaborate with internal and external research teams** to integrate bioinformatic pipelines into ecological and evolutionary immunology studies, particularly focusing on symbiosis biology.
- **Contribute to proposal writing and strategic planning** for the research center, assisting in defining research objectives, outreach initiatives, and assessment strategies.
- **Facilitate interdisciplinary partnerships**, bridging computational biology with ecology to enhance research capacity and promote long-term institutional development in bioinformatics.

Technical Skills:

- **Bioinformatics & Genomics:** 16S rRNA gene analysis, RNA-Seq library preparation, Illumina sequencing (NGS), metagenomic data processing
- **Computational Tools:** R, Python, Bash scripting, GitHub, high-performance computing (HPC), command-line interface (Unix/Linux)
- **Data Science & Visualization:** Statistical modeling, data visualization (ggplot2, Cytoscape), multivariate ecological analyses
- **Professional & Research Skills:** Experimental design, scientific writing, presentation development, teaching & mentorship, interdisciplinary collaboration

Graduate Assistant 2019-2024

University of Florida – Department of Soil, Water, and Ecosystem Sciences
Advisor: Dr. Julie Meyer

- **Led independent research** investigating the role of coral microbiomes in promoting coral health and resilience under marine deoxygenation stress.
- **Designed and conducted molecular experiments**, integrating high-throughput sequencing, microbial community analysis, and bioinformatics to assess coral-associated microbial dynamics.

- **Performed extensive field sampling** across diverse marine environments, collecting and processing coral and water samples for genomic and microbiome analyses.
- **Developed and optimized bioinformatic pipelines** for 16S rRNA gene and RNA-Seq data analysis, employing high-performance computing (HPC) for large-scale sequence processing.
- **Executed molecular laboratory workflows**, including DNA/RNA extraction, Covaris fragmentation, PCR, and library preparation for Illumina sequencing.
- **Applied statistical and computational methods** in R and Python to analyze sequencing data, visualize microbial community shifts, and interpret functional genomic patterns.
- **Mentored undergraduate students**, providing guidance in molecular techniques, data analysis, and research methodologies.
- **Presented research findings** at national and international conferences, contributing to the broader scientific understanding of microbial ecology in marine ecosystems.

Technical Skills:

- **Molecular & Microbial Techniques:** DNA/RNA extraction, PCR, 16S rRNA gene sequencing, RNA-Seq, epifluorescence microscopy, Covaris fragmentation
- **Bioinformatics & Computational Biology:** R, Python, Bash scripting, GitHub, HPC job management, command-line interface, multivariate ecological statistics
- **NGS & Data Analysis:** Illumina sequencing, microbial community profiling, functional gene analysis, data visualization (ggplot2)
- **Fieldwork & Experimental Design:** Marine sample collection, AAUS scientific diving, water quality monitoring, coral health assessments
- **Professional & Research Skills:** Scientific writing, presentation development, collaboration, troubleshooting experimental workflows

Rhode Island INBRE Fellow 2018-2019

Roger Williams University

Advisor: Dr. Koty Sharp

- **Conducted research** on microbial dynamics associated with microplastics ingestion in the coral *Astrangia poculata*, assessing microbial shifts and potential ecological impacts.
- **Designed and executed experiments** to investigate microplastic-associated microbial communities, integrating field sampling with controlled laboratory studies.
- **Performed microscopy-based analyses** and image documentation to assess coral-microbe interactions and microplastic retention within coral tissues.
- **Maintained and monitored** coral husbandry systems, ensuring optimal water quality and environmental conditions for experimental trials.
- **Collaborated with faculty and peers** to refine methodologies and present findings in research symposiums and conferences.

Technical Skills:

- **Laboratory & Experimental Methods:** Sample preparation, microscopy, image analysis, microplastic handling
- **Field & Aquatic Systems:** Coral husbandry, water quality monitoring, experimental tank maintenance
- **Research & Communication:** Experimental design, data documentation, scientific presentations, collaboration

Student Researcher August 2018-December 2018

Bermuda Institute of Ocean Sciences

- **Conducted independent research** to assess the thermotolerance of the coral *Diploria labyrinthiformis* and its algal symbionts (Symbiodiniaceae) across a depth gradient.

- **Utilized advanced microscopy techniques** and hemocytometer assays to quantify algal symbiont densities and assess cellular health.
- **Applied spectrometric analysis** to evaluate chlorophyll concentration and other biochemical markers of thermal stress in both coral and symbiont tissues.
- **Participated in field sampling and AAUS scientific diving** to collect samples from for laboratory analysis.

Technical Skills:

- **Experimental Design & Laboratory Techniques:** Microscopy, hemocytometer, spectrometry, coral stress assays
- **Aquatic Systems & Monitoring:** Aquaculture, water quality management
- **Fieldwork & Diving:** AAUS scientific diving, coral collection, environmental data logging
- **Research & Communication:** Data analysis, scientific writing, experimental documentation, scientific presentations

Research Assistant

January 2018-May 2018

Roger Williams University, Wet Laboratory

Advisor: Dr. Casey Murray.

- **Collaborated on an aquaculture nutrition project** investigating the effectiveness of alternative protein replacements in fish feed, aiming to optimize feed efficiency and sustainability in aquaculture.
- **Conducted feeding trials**, monitoring fish growth and overall health in response to various protein formulations.
- **Managed fish husbandry** including daily care, feeding schedules, and health monitoring to ensure optimal conditions for experimental accuracy.
- **Monitored and maintained water quality** throughout the study, ensuring appropriate environmental parameters for fish survival and growth.
- **Utilized ImageJ** for image analysis and quantification of fish body condition and other growth metrics.
- **Implemented humane euthanasia techniques** in compliance with ethical guidelines for research involving live animals.
- **Collaborated closely with team members** to analyze data and develop recommendations for future feed formulation studies.

Technical Skills:

- **Aquaculture & Fish Husbandry:** Fish care, feeding protocols, growth monitoring
- **Laboratory & Experimental Techniques:** Nutritional trials, euthanasia techniques, ImageJ analysis
- **Environmental Monitoring:** Water quality assessment, tank manipulation
- **Research & Collaboration:** Data analysis, teamwork, experimental documentation

PUBLICATIONS

PEER REVIEWED PRIMARY RESEARCH PUBLICATIONS

2023

Howard RD, Schul MD, Rodriguez Bravo LM, Altieri AH, Meyer JL (2023). Shifts in the coral microbiome in response to *in situ* experimental deoxygenation. *Applied and Environmental Microbiology* 89(11): e0057723. doi.org/10.1128/aem.00577-23

PEER REVIEWED PRIMARY RESEARCH PUBLICATIONS IN PREPARATION

Howard RD, Beavers K, Huntley N, Rossin AM, Apprill A, Holstein DM, Correa AMS, Brandt M, Mydlarz LD**.** Panomics reveals coordinated holobiont signatures of stony coral tissue loss disease (SCTLD) through supervised machine learning (ML) methods.

Oviedo L, Borbee E, **Howard RD**, Fuess L. Integrative analysis of mixed symbiosis in *Astrangia poculata*.

Howard RD, Schul MD, Lucey NM, Januszkiewicz KE, Collin R, Altieri AH, Meyer JL. Examining the impact of historical hypoxia on coral microbial communities across Bahía Almirante, Panama.

Howard RD, **Quintanilla E**, Lucey NM, Januszkiewicz KE, Collin R, Altieri AH, Meyer JL. Differential gene expression in the microbiome of the massive starlet coral *Siderastrea siderea* under deoxygenation.

PRESENTATIONS

Rachel Howard, Kelsey Beavers, Naomi Huntley, Ashley Rossin, Amy Apprill, Daniel Holstein, Adrienne Correa, Marilyn Brandt, Laura Mydlarz. Panomics reveals coordinated holobiont signatures of stony coral tissue loss disease (SCTLD) through supervised machine learning (ML) methods. Southeast Texas Evolutionary Genetics & Genomics Symposium 2026 (poster presentation). May 2026.

Rachel Howard, Centers for Research Excellence in Science and Technology (CREST) Education & Training Workshop. April 2025.

Rachel Howard, Elena Quintanilla, Monica Schul, Andrew Altieri, Julie Meyer. Hypoxia and the coral microbiome: How the coral microbiome may impact host health and resilience in response to an unexplored tropical stressor. 2023 North Florida Marine Science Symposium (oral presentation). March 2023.

Rachel Howard, Monica Schul, Andrew Altieri, Julie Meyer. Shifts in the coral microbiome in response to *in situ* experimental deoxygenation. 20th Annual Soil, Water, and Ecosystem Research Forum (oral presentation). February 2023.

Rachel Howard, Monica Schul, Andrew Altieri, Julie Meyer. Shifts in the coral microbiome in response to *in situ* experimental deoxygenation. Center for Coastal Solutions State of the Center Summit (poster presentation). December 2022.

Rachel Howard, Monica Schul, Andrew Altieri, Julie Meyer. Shifts in the coral microbiome in response to *in situ* experimental deoxygenation. Florida Genetics Symposium 2022 (poster presentation). November 2022.

Rachel Howard, Monica Schul, Andrew Altieri, Julie Meyer. Shifts in the coral microbiome in response to *in situ* experimental deoxygenation. 15th International Coral Reef Symposium (oral presentation). July 2022.

Rachel Howard, Monica Schul, Andrew Altieri, Julie Meyer. Shifts in the coral microbiome in response to *in situ* experimental deoxygenation. American Society of Microbiology Microbe 2022 (poster presentation). June 2022.

Rachel Howard, Monica Schul, Andrew Altieri, Julie Meyer. Shifts in the coral microbiome in response to *in situ* experimental deoxygenation. Florida Branch of the American Society for Microbiology (oral presentation). February 2022.

Rachel Howard, Monica Schul, Andrew Altieri, Julie Meyer. Shifts in the coral microbiome in response to *in situ* experimental deoxygenation. International Coral Reef Symposium, Virtual conference (oral presentation). July 2021.

Rachel Howard. Optimizing metagenomic libraries for coral microbiome studies. Graduate Student Showcase, University of Florida Genetics Institute (oral presentation). April 2021.

Rachel Howard, Leah Hintz, Allison Klein, Alicia Schickle, Randi Rotjan, and Koty Sharp. Investigating the plastisphere: the role of plastic-associated microbes on microbead ingestion by the coral *Astrangia*

poculata. *Astrangia* Research Workshop, Roger Williams University (oral presentation). May 2019.

Rachel Howard. Depth-dependent bleaching: thermotolerance of *Diploria labyrinthiformis* and its symbionts along a depth gradient. Bermuda Institute of Ocean Sciences (oral presentation). November 2018.

Leah Hintz, **Rachel Howard**, Allison Klein, Alicia Schickle, Randi Rotjan, and Koty Sharp. Investigating the plastisphere: The role of plastic-associated microbes on microbead ingestion by the coral *Astrangia poculata*. Rhode Island SURF Undergraduate Research Symposium, University of Rhode Island (poster presentation). July 2018.

Teaching Appointments

Primary Instructor — Spring 2025
Texas State University – Department of Biology
Fundamentals of Bioinformatics Workshop

Teaching Assistant — Spring 2020, 2021, 2022, 2023
University of Florida – Department of Soil, Water, and Ecosystem Sciences
SWS 4307/5308 – Ecology of Waterborne Pathogens

Teaching Assistant — Summer 2019, 2020
University of New Hampshire & Cornell University – Shoals Marine Laboratory
BIOSM 1620 – Marine Environmental Science

Science Tutor — Spring 2017, 2018, 2019
Roger Williams University – The Tutoring Center – Science Center

Professional Service & Development

TEACHING DEVELOPMENT

UF Preparing Future Faculty Program — Fall 2023

PUBLIC OUTREACH

UF NRDI (National Resource Diversity Initiative) in partnership with the Cultural Arts Coalition presents afterschool science activities to K-8th students:

Resilience Charter School, Gainesville, FL (2023)
George Washington Carver Group, Gainesville, FL (2023) Caring & Sharing Learning School, Gainesville, FL (2023)

Skype a Scientist interview with Girl Guides: 1st Forest Girl Guide Unit, Ontario, Canada (2021)

Scientist in Every Florida School Interview with high school students, Fort Pierce, FL (2021) Skype a Scientist interview with 7th and 8th grade students, Lakeside, CA (2021)

Skype a Scientist interview with 3rd grade students, Swanville, ME (2020)

AFFILIATIONS & OTHER ACTIVITIES

American Academy of Underwater Sciences (AAUS) — 2018-Present
Science diver

NAUI Nitrox Diver 2023-Present

Awards & Honors

Invited Speaker – Soil, Water and Ecosystem Sciences February 2023
University of Florida

Institute of Food and Agricultural Sciences Travel Grant February 2022
University of Florida

Latin Honors – *Magna Cum Laude* May 2019
Roger Williams University

Best Undergraduate Presentation May 2019
Astrangia Research Workshop

Dean's List Spring 2017, Fall 2018, Fall 2019, Spring 2019
Roger Williams University

Presidential Scholarship 2017-2019
Roger Williams University

Rhode Island IDeA Network of Biomedical Research Excellence (RI-INBRE) Fellowship 2018- 2019

Summer Undergraduate Research Fellowship (SURF) Summer 2018

Field Expeditions

Field Work: Smithsonian Tropical Research Institute, Panama July-August 2022
Collect coral samples via SCUBA to study impact of hypoxia on the coral microbiome. Fish surveys via SCUBA. Ran tank experiments to simulate hypoxia to study gene expression shifts of the coral microbiome.

Field Work: Central Caribbean Marine Institute, Cayman Islands January 2020
Collect coral samples via SCUBA to study the microbiome pre-Stony Coral Tissue Loss Disease

Study Abroad: Bermuda Institute of Ocean Sciences, Bermuda Fall 2018
Roger Williams University, Bermuda Institute of Ocean Sciences
AAUS Science Diving in: Coral Reef Ecology (BIO361), Marine Biological Oceanographic Research (BIO410), Research Diving Methods (BIO410)

Field Work: Rhode Island Summer 2018
Deploying & retrieving gear to research microbe colonization on microplastics

Study Abroad: Belize Winter Intersession 2018
Roger Williams University
Tropical Ecology (BIO310)